IOT-BASED SMART ATTENDANCE SYSTEM USING RFID

Mini Project Report

IN

Master of Computer Application

Submitted by

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1. ABSTRACT

This project presents an IoT-Based Smart Attendance System using RFID technology to automate the attendance process. Traditional methods are time-consuming and prone to errors, whereas this system provides a fast and reliable solution.

The system uses a NodeMCU (ESP8266) and RFID module (MFRC522) to read the unique ID of RFID cards. The data is sent to a server using IoT communication over WiFi, where attendance is recorded in real time.

It reduces manual effort, improves accuracy, and provides instant feedback using LCD, buzzer, and LED. The system is efficient, secure, and suitable for modern digital environments.

2. INTRODUCTION

Attendance management is an important task in schools and organizations, but traditional methods are slow and error-prone. With the help of Internet of Things (IoT), attendance systems can be automated and made more efficient.

This project develops an IoT-Based Smart Attendance System using RFID, where each user is identified using an RFID card. The RFID reader reads the card and sends the data to the server through NodeMCU using WiFi.

The system records attendance automatically, provides real-time updates, and reduces human errors. It is a smart, reliable, and easy-to-use solution for modern attendance management.

3. OBJECTIVES

The main objectives of the project are:

- To automate attendance using RFID and IoT
- To enable real-time data transmission using WiFi
- To reduce manual errors and proxy attendance
- To store data in a centralized database
- To provide real-time monitoring and analytics
- To develop a scalable and secure system

4. SYSTEM OVERVIEW

The proposed system is based on IoT architecture and consists of three main components:

1. IoT Device Layer: This layer includes the hardware components such as NodeMCU (ESP8266), RFID reader (MFRC522), LCD display, buzzer, and LED. It is responsible for capturing user data through RFID card scanning and providing real-time feedback.

2. Communication Layer: This layer enables data transmission using WiFi (ESP8266) and HTTP protocol. It connects the hardware device to the server and ensures real-time communication.

3. Cloud/Server Layer: This layer consists of the backend server (Node.js) and database (MongoDB). It processes the received data, stores attendance records, and manages system operations.

IoT Data Flow: RFID Card → NodeMCU → WiFi → Server → Database → Dashboard

This real-time data flow is the key feature of the IoT-based system, ensuring instant attendance recording and monitoring.

5. HARDWARE COMPONENTS

5.1 NodeMCU (ESP8266)

Definition: NodeMCU is an open-source IoT development board based on the ESP8266 microcontroller, which has built-in WiFi capability for wireless communication. The ESP8266 is a low-cost microcontroller with an integrated TCP/IP protocol stack, which enables devices to connect directly to the internet. It acts as the brain of the IoT system, processing input data and transmitting it to the server.

In this project, NodeMCU:

- Reads data from the RFID module
- Processes the UID
- Sends data to the backend server via WiFi using HTTP

It eliminates the need for external WiFi modules, making it ideal for IoT applications.

5.2 RFID Module (MFRC522)

Definition: RFID (Radio Frequency Identification) is a wireless technology used for automatic identification of objects using radio waves. The MFRC522 is a commonly used RFID reader module. RFID works on the principle of electromagnetic fields. The RFID

reader generates radio waves that activate the RFID tag, which then transmits its stored data (UID) back to the reader.

The MFRC522 module uses:

- SPI communication protocol to communicate with NodeMCU
- Short-range communication (2–5 cm)

In this system:

- The RFID reader scans the card
- Extracts the unique ID (UID)
- Sends it to NodeMCU for processing

5.3 RFID Cards/Tags

Definition: RFID cards are passive electronic devices that store a unique identification number (UID) and communicate with RFID readers using radio waves. RFID cards are typically passive tags, meaning they do not have their own power source. They receive energy from the electromagnetic field generated by the RFID reader.

When the card is brought near the reader:

- It gets activated
- Sends its UID back to the reader

Each card has a unique UID, which is used in this system to identify users and mark attendance.

5.4 LCD Display (16x2)

Definition: A 16x2 LCD is an electronic display module capable of displaying 16 characters per line across 2 lines. The LCD operates using liquid crystal technology, where the alignment of liquid crystals changes when voltage is applied, allowing characters to be displayed.

It usually communicates using:

- I2C protocol (in this project for fewer wires)
- In this system, the LCD:
- Displays messages like “Scan Card”
- Shows attendance status (IN/OUT)
- Provides real-time feedback to the user

5.5 Buzzer

Definition: A buzzer is an audio signaling device that produces sound when an electrical signal is applied.The buzzer works on the principle of converting electrical energy into sound energy using vibration.

In this project:

- Short beep → Successful scan
- Long beep → Error or invalid card

It enhances user interaction by providing immediate audio feedback.

5.6 LED (Light Emitting Diode)

Definition: An LED is a semiconductor device that emits light when current flows through it. LEDs work on the principle of electroluminescence, where electrons recombine with holes and release energy in the form of light.

In this system:

- Green LED → Successful attendance
- Red LED → Error or invalid scan

LEDs provide quick visual feedback to users.

6. SOFTWARE & TECHNOLOGIES

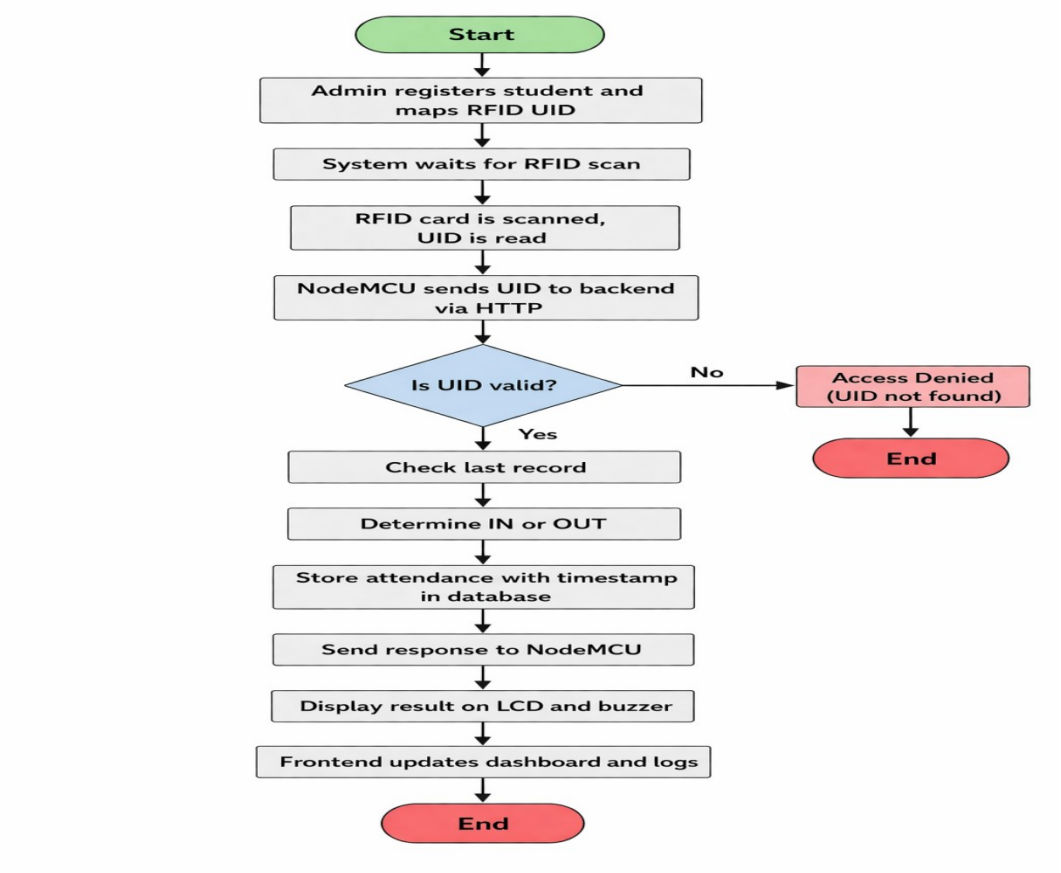
The system uses a combination of software tools and technologies to ensure smooth operation and real-time communication:

- Embedded Programming: Arduino IDE is used to program the NodeMCU (ESP8266). It controls the RFID module, reads the UID of cards, and sends data to the server.
- Backend: Node.js and Express are used to build the server-side application. They handle API requests, process attendance logic, and manage communication with the database.
- Database: MongoDB is used as a NoSQL database to store user details and attendance records in a structured format.
- Frontend: React is used to develop an interactive and user-friendly dashboard for viewing attendance data, logs, and analytics.
- Communication: HTTP protocol is used for communication between the IoT device and the server, enabling data transfer over WiFi.

7. WORKING PRINCIPLE

- The working of the system is carried out step-by-step as follows:
- The user scans the RFID card near the RFID reader.
- The RFID reader reads the unique ID (UID) of the card.
- The NodeMCU processes the UID received from the reader.
- The NodeMCU connects to the WiFi network.
- The UID is sent to the backend server using an HTTP request.
- The server validates the UID with the stored database.
- The system determines whether the attendance is IN (entry) or OUT (exit).
- The attendance data is stored in the database with date and time.
- A response is sent back to the NodeMCU.
- The LCD displays the attendance status, and buzzer/LED provide feedback.
- This entire process is performed in real-time using IoT communication, ensuring fast and accurate attendance recording.

Fig 7.1: System Workflow Algorithm



8. HARDWARE IMPLEMENTATION (PIN CONNECTION)

This section describes the circuit connections and implementation of hardware components used in the IoT-Based Smart Attendance System.

1. Circuit Connections

1.1 RFID Module (MFRC522)

RFID Pin	NodeMCU Pin
VCC	3.3V
GND	GND
SDA (SS)	D8
SCK	D5
MOSI	D7
MISO	D6
RST	D3

1.2 LCD Display (I2C) Connections

LCD Pin	NodeMCU Pin
VCC	3.3V
GND	GND
SDA	D2
SCL	D1

1.3 LED & Buzzer Connections

LED Pin	NodeMCU Pin	Buzzer Pin	NodeMCU Pin
Positive (+)	D0	Positive (+)	D4
GND	GND	GND	GND

2. Implementation Overview

- RFID reader scans the card and sends UID to NodeMCU
- NodeMCU processes the data and sends it to the server

- Based on server response, LCD displays result
- LED and buzzer provide instant feedback

3. Key Points

- RFID module operates on 3.3V only
- SPI pins must be connected correctly
- Stable WiFi is required for IoT communication

9. OUTPUT/ SCREENSHOTS

Fig 13.1: RFID Attendance System Component Connection

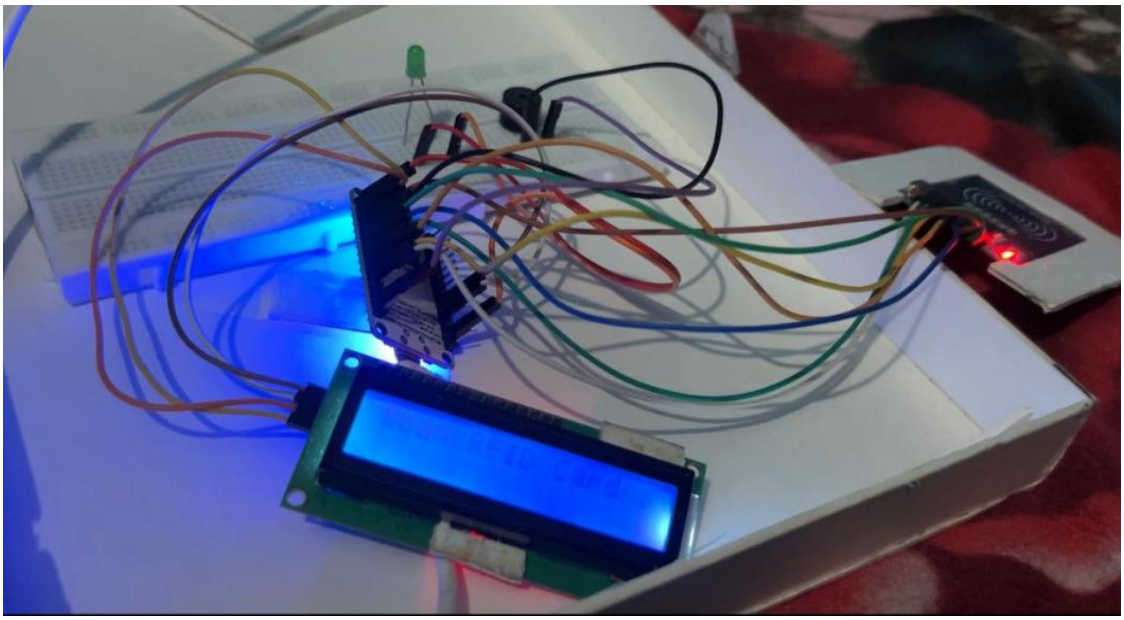
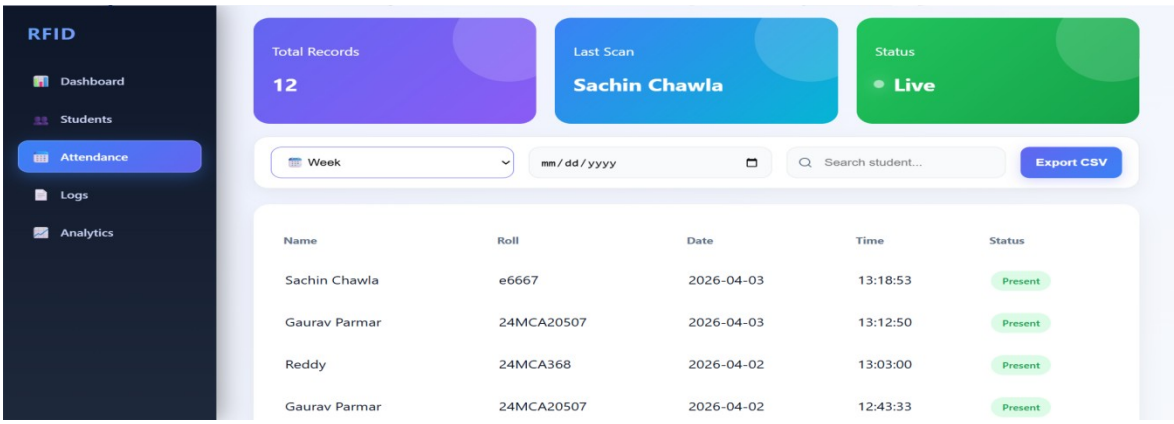


Fig 13.2: LCD Display Output

Fig 13.3: Web Dashboard Interface



10. ADVANTAGES OF IOT-BASED SYSTEM

- Real-time attendance tracking
- Reduced manual effort
- High accuracy
- Centralized data storage
- Easy monitoring from anywhere
- Scalable system
- Faster processing

Compared to traditional systems, IoT enables instant updates and remote access .

10. LIMITATIONS

- Requires stable internet connection
- RFID cards can be shared (proxy issue)
- Limited RFID range

11. APPLICATIONS

- Schools and colleges
- Offices and companies
- Laboratories
- Secure areas

12. FUTURE SCOPE

- Integration with biometric systems
- Mobile app development
- Cloud storage integration
- Face recognition system
- Real-time notifications

13. CONCLUSION

The IoT-Based Smart Attendance System provides an efficient and automated solution for attendance management. By integrating RFID technology with IoT communication, the system ensures real-time data processing, improved accuracy, and reduced manual effort.

The project demonstrates how IoT can transform traditional systems into smart and intelligent solutions.